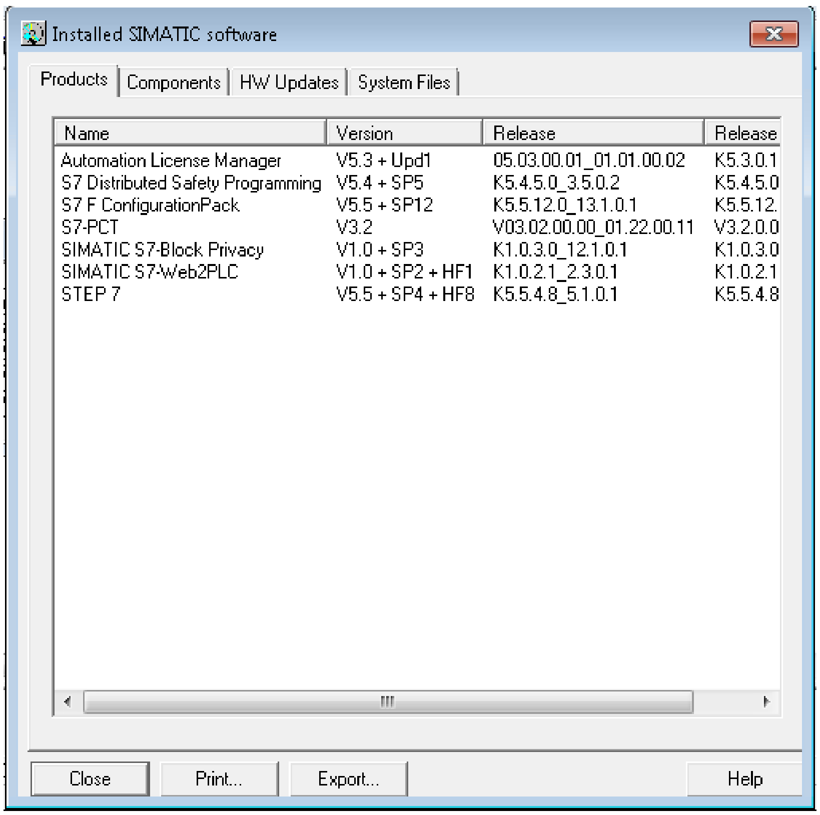
ProfiNet Readme (v0.3)

1. STEP 7 Requirement

STEP 7 V5.5 is necessary in order to have IO-Device and IO-Controller working together (same channel). S7 Distributed Safety Programming (V5.4+ SP5 or above) and F Configuration pack are necessary package in order to make safety work.



2. Network Configuration in CP1604

Setting up IP address, subnet mask, and device name of CP1604

- Start up R-30iB in SUSPEND mode
- These settings are important and should be matched to those used in creating STEP7 Project.
- In our experience, mismatch of device name is the most likely in configuration error.

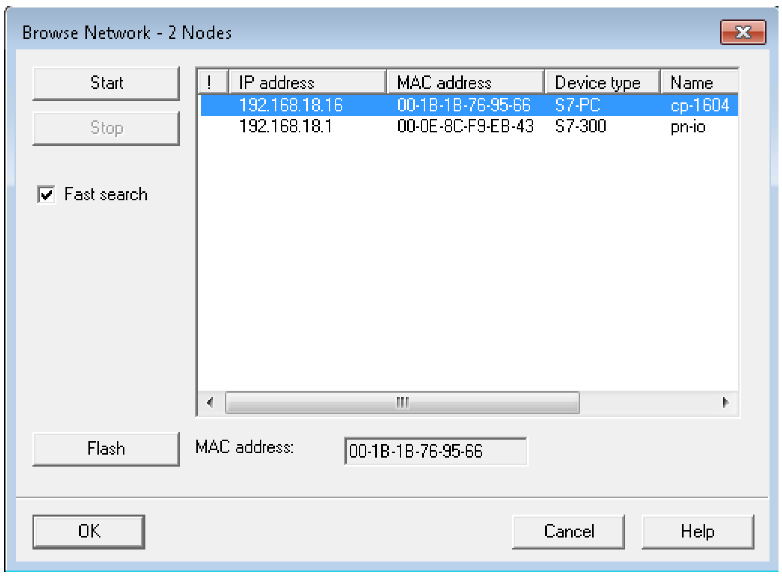
NOTE:

- IP address must be set beforehand. “Assign IP address via IO Controller” can’t be used. If IP address is not set, PRIO-605 "PNIO(D) open error 030D" shows up.

- Device name cannot be modified if CP1604 has been configured as I/O Controller. In this case, press "Reset" button in the "Reset to factory setting" field to initialize CP1604.

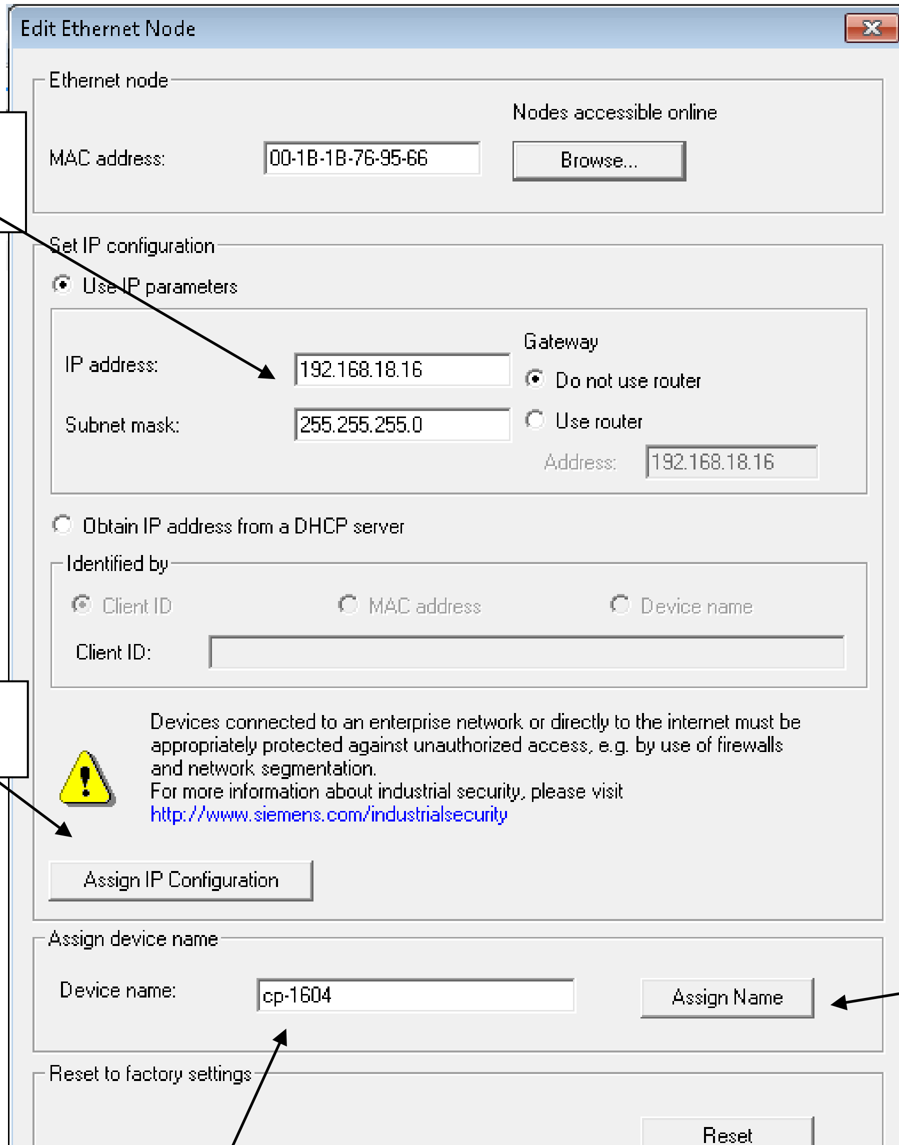
Procedure (STEP 7):

- Start STEP 7 and in "SIMATIC Manager" from start menu, for example.
- Click "Edit Ethernet Node" from "PLC" tool bar.
- Press "Browse" button. The configuration tool will search all devices connected to the Ethernet. The window like below appears after a while.
- Locate device type S7- PC or S7-300 and select it. Then press "OK".



Note: *I believe this field changes based on whether it is device only or device/controller and the connection status of the device.*

- The window like below appears. IP address and device name can be set from this window.
- Enter IP address and subnet mask. Press "Assign IP Configuration".
- Enter device name and press "Assign Name" button.
- This procedure is completed. Press "Close" button to close the window.



A. Enter IP address and subnet mask.

B Press Assign IP Config

C. Enter Device name and leave this blank if configuring IO-Controller

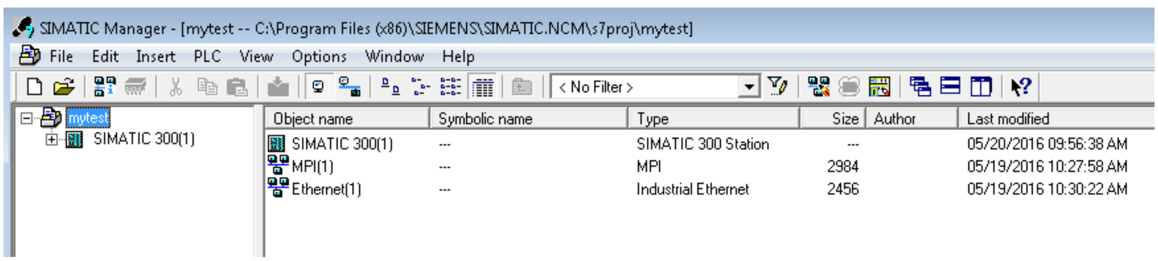
D. Press Assign Name

3. IO-Device Configuration

Safety module MUST be first after head module. See below. Safety is configured for 2 bytes and IO is 16/16 bytes. Robot is IO-Device and PLC is IO-Controller.

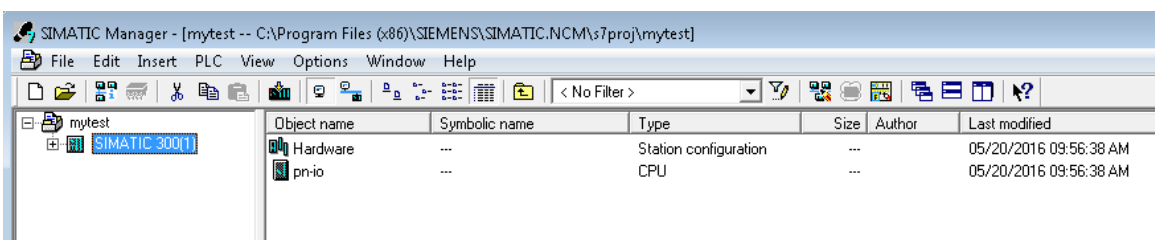
Procedure (STEP 7): Create a project

- Start “SIMATIC Manager” from start menu, for example.
- Open “File” tool bar and select “New”.
- Enter Project Name. Press OK.
- Right Click the project window.
- Insert New Object->Industrial Ethernet. The default name is Ethernet(1).
- Insert New Object->SIMATIC 300 Station. The default name is SIMATIC 300(1).



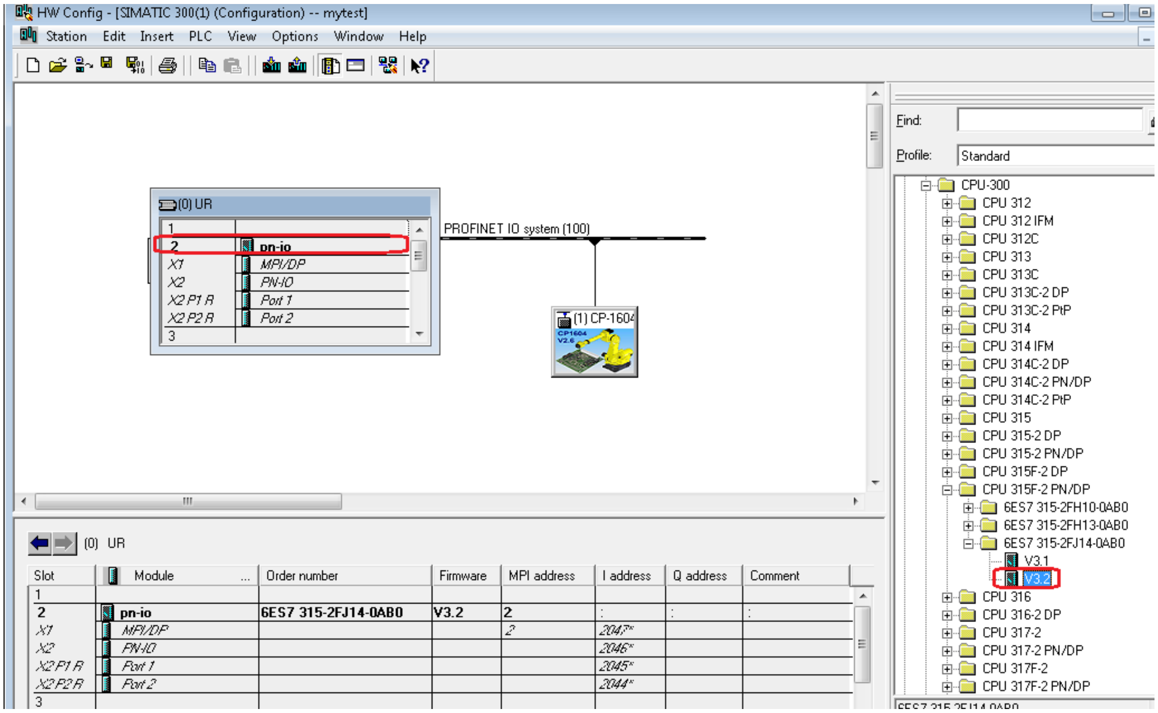
Procedure (STEP 7): Enter PLC Configuration.

- Click “SIMATIC 300(1)” icon in the project window (left).
- Double click “Hardware” in the project window (right).
- HW Config window will show up.

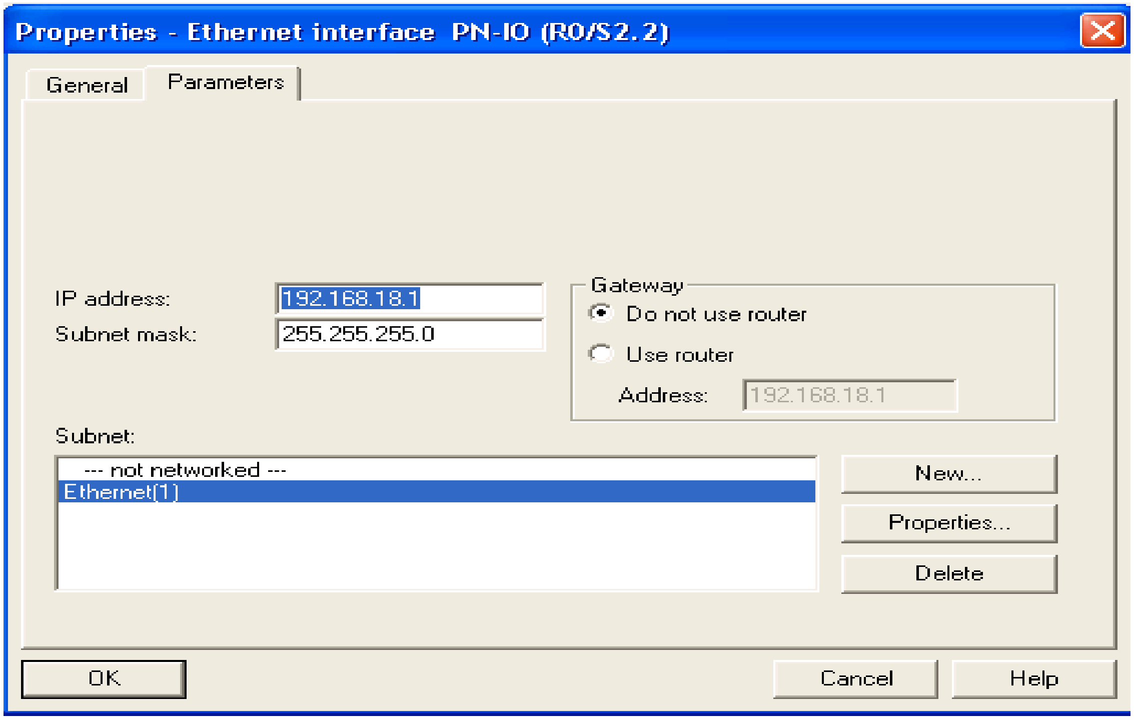


- There is a list in the right side of HW Config window. Move there and search SIMATIC 300->Rack-300->Rail and double click, and then small window with many slots appears.
- Select slot2. Double click the following.

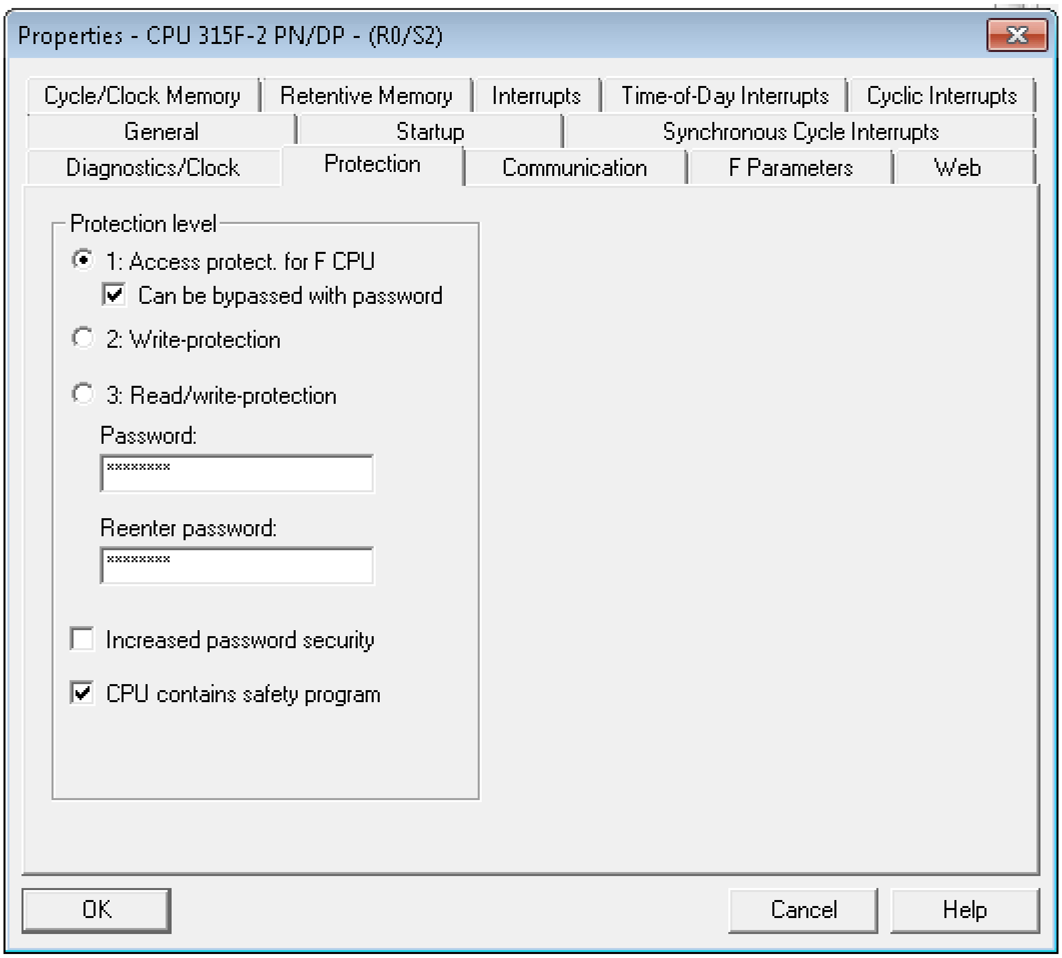
SIMATIC 300->CPU315->CPU315F-2PN/DP->6ES7-2FJ14-0AB0->V2.6 (for example). You have to choose correct CPU module instead of that in the example.



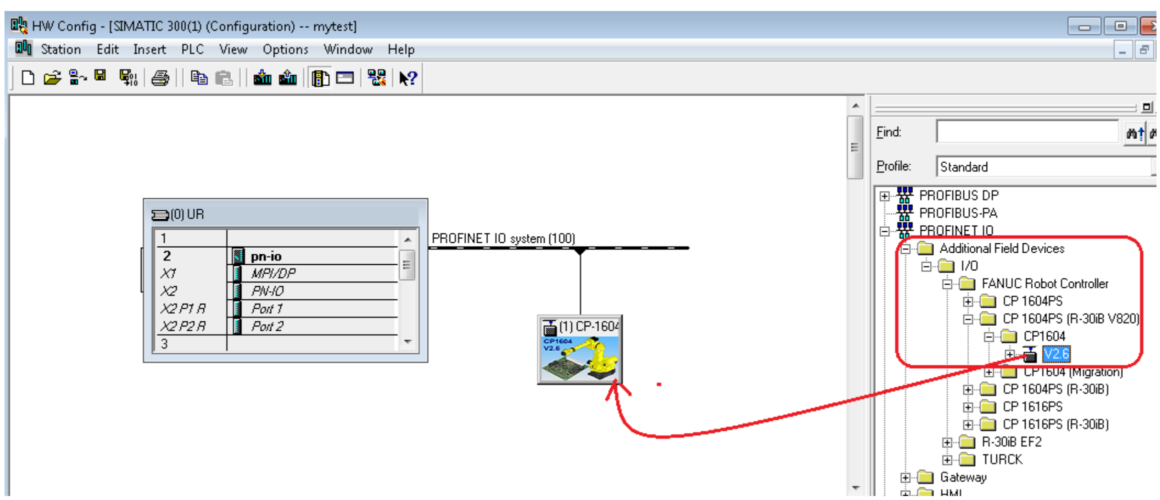
- New window appears.
    - Enter IP address and subnet mask of PLC.
    - Select “Ethernet(1)” as subnet.
    - Press OK. The window is closed.
- Select “Save and Compile” from menu. This will show up in RAIL in the left but with name of model number which you can rename it whatever you name you gave to your PLC. In this case, *pn-io*.



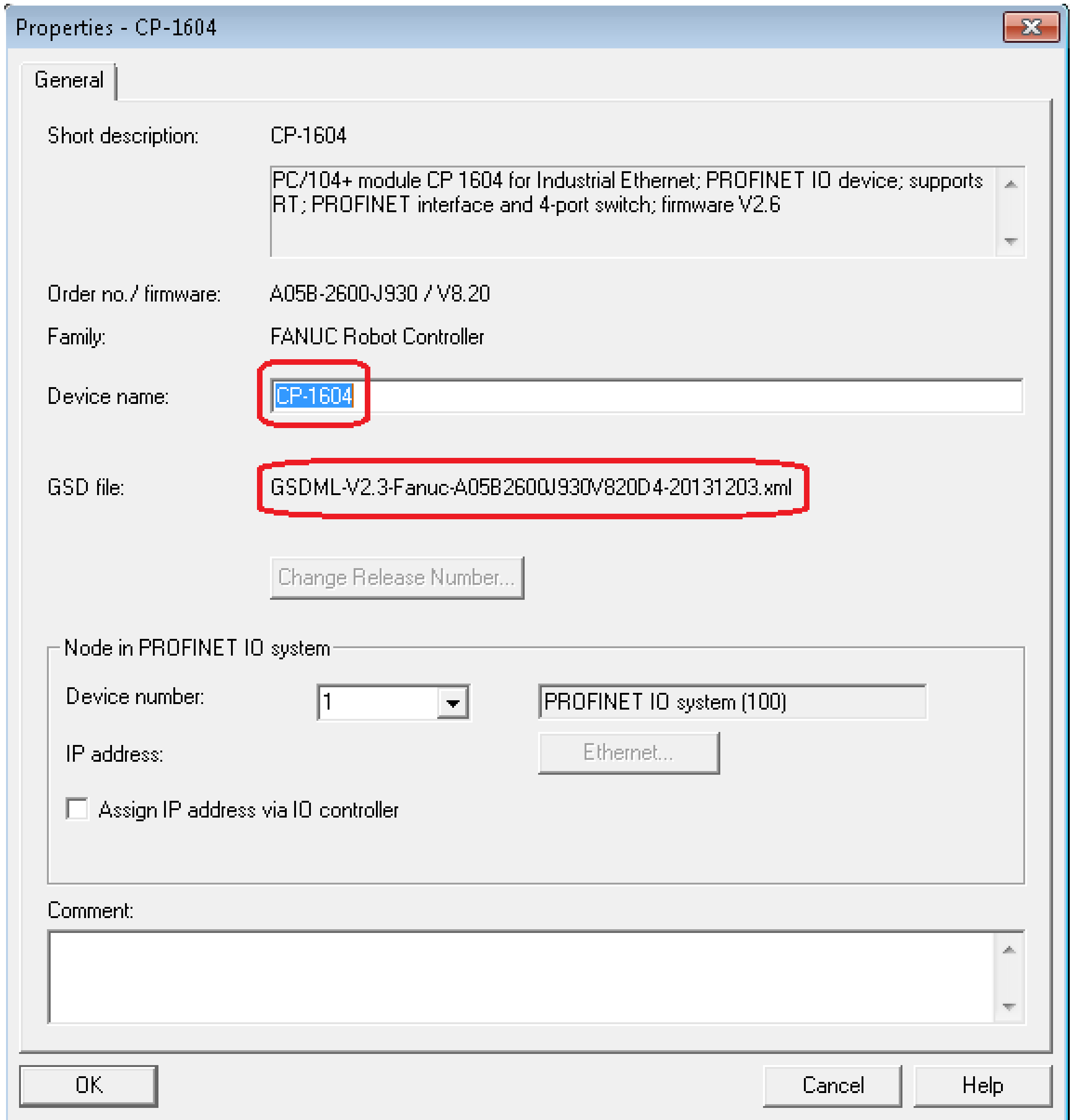
- Click Properties of CPU (slot 2 in RAIL) and Select Protection TAB. Enter Password and check "*Can be bypassed with password*" and "*CPU contains safety Program*".



Drag V2.6 from the right pane and drop on the PROFINET IO sysem.



Now double click on CP-1604 icon. You will see below.

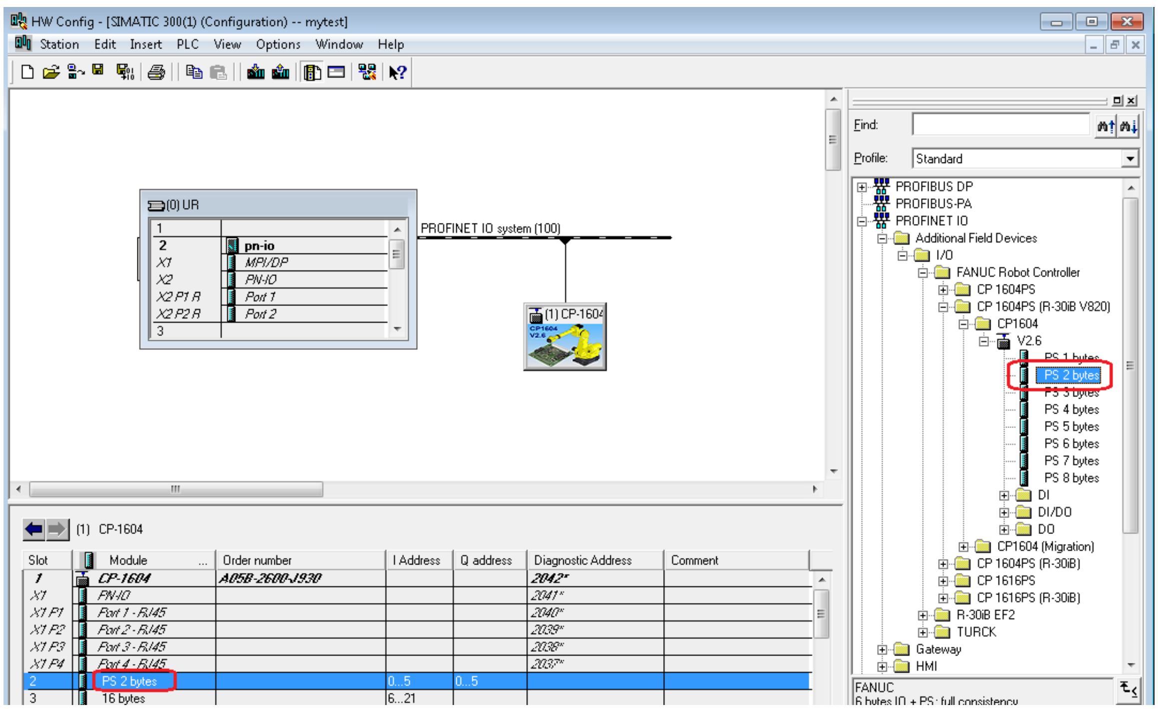


- Make sure device name matches to that set to CP1604 card during initial step.
- For your information, the name of GSDML file is displayed in above.
- Select device number usually 1 if not other devices configured.
- Remove the check in “Assign IP address via IO Controller”. You may not be able to do it in Molex Profinet EF2 IOD. That is not an issue. It is often grayed out in EF2 IOD.
- Now “Save and Compile” button from the menu.

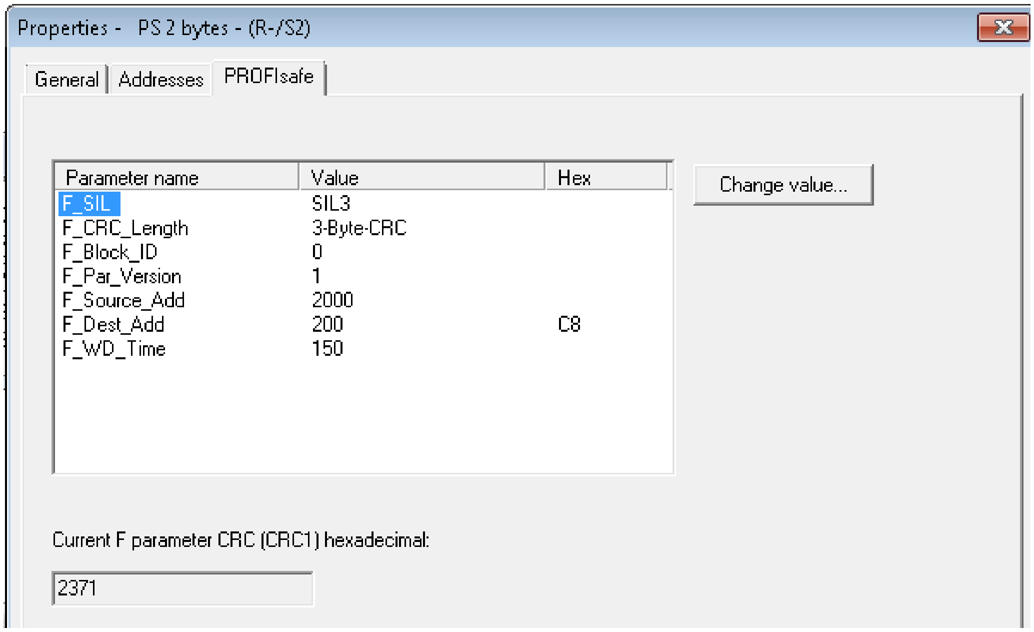
Click on CP1604 module and Slot assignment will show up in HW Config window in lower sub-window.

Procedure (STEP 7): Configure Safety Module

Drag "PS 2 Bytes" from right pane to slot #2 at bottom left pane. Add standard I/O for I/O device. In this example, 16 Bytes DI and 16 Bytes DO in same order. Just drag appropriate DI/DO from right like Safety I/O. Now double click on the newly added safety I/O.



Go to PROFISafe tab. This screen tells you safety related information. Most important item is *F_Dest_Add* which MUST match in robot controller.

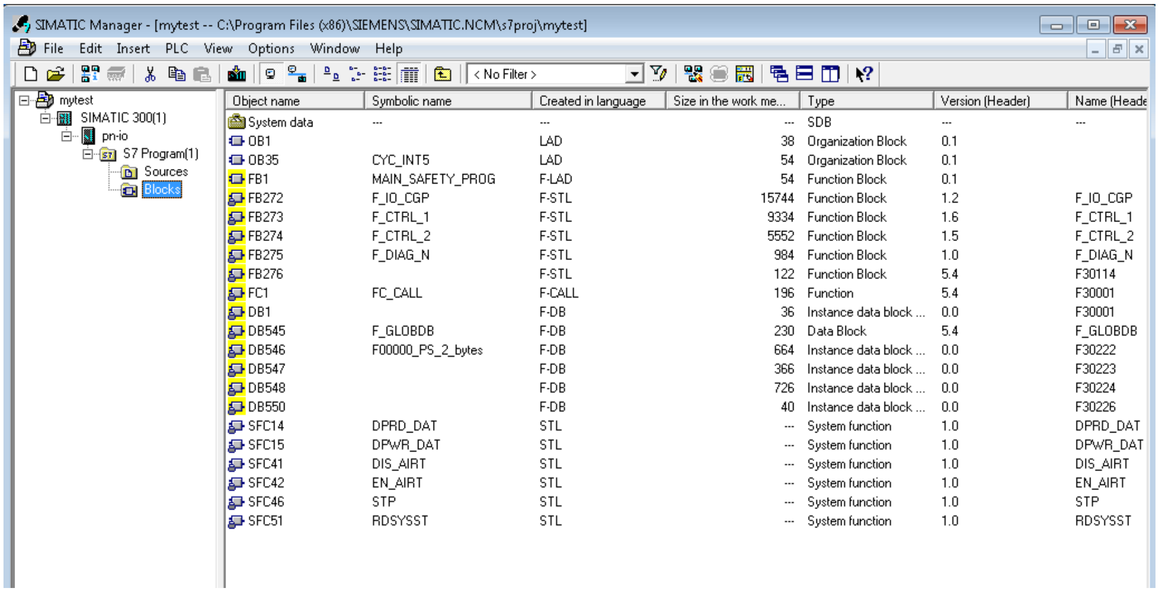


Now create Function call (FC), Function block (FB), Data block (DB) and OB (Object Block) for example FC1, FB1, DB1 and OB35 using below procedure.

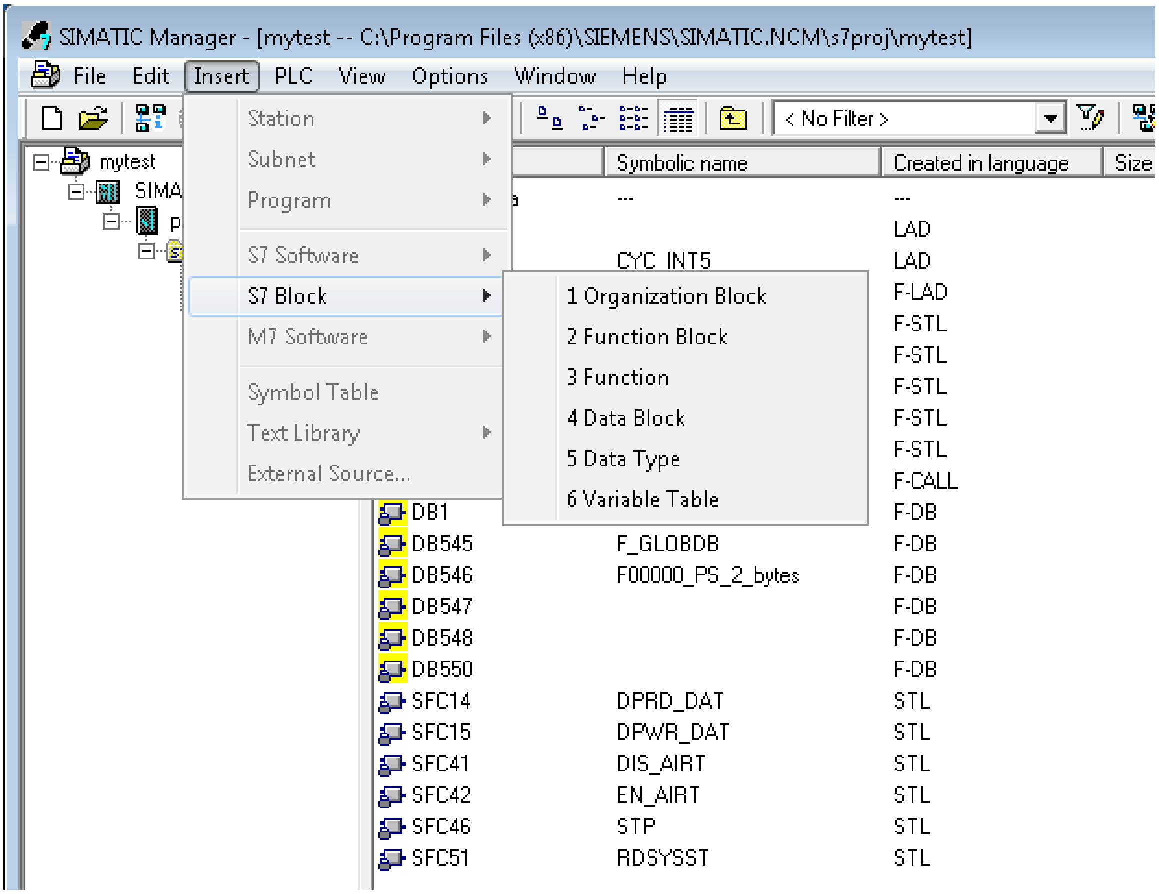
Block Name	**Block Language**	**Description**
Function (FC)	F-CALL	F-block for calling the F-runtime group from the standard user program. The F-CALL contains the call for the F-program block and the calls for the automatically added F-blocks of the F-runtime group. You create the F-CALL, but you cannot edit it. It is possible to call the F-CALL in an OB or FB/FC that is called in an OB.
Function Block (FB)	F-LAD	You program the actual safety function using F-FBD or F-LAD. The starting point for F-programming is the F-program block. The F-PB is an F-FC or F-FB (with instance DB) that becomes the F-PB when assigned to the F-CALL.
Data Block (DB)	F-DB	It's language should be F-DB and Instance of FB40
Object Block (OB)	LAD	Requires F-CALL to be added in ladder. 1. Double Click on OBxx and look at left tree. 2. It MUST have FC40 created recently. 3. Drag it to right pane in ladder.

		4. Save and Close

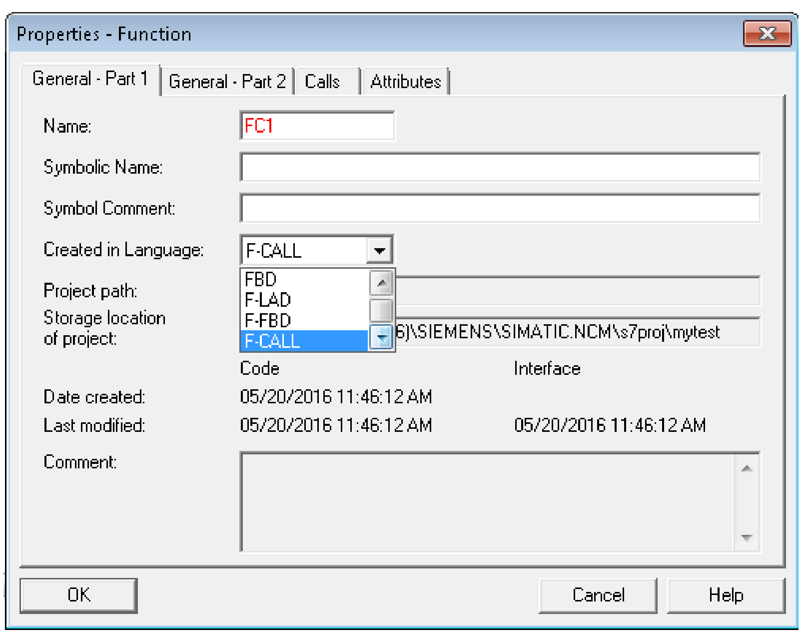
In order create all these blocks. Go to SEMATIC Manager navigate to **Blocks** in left pane.



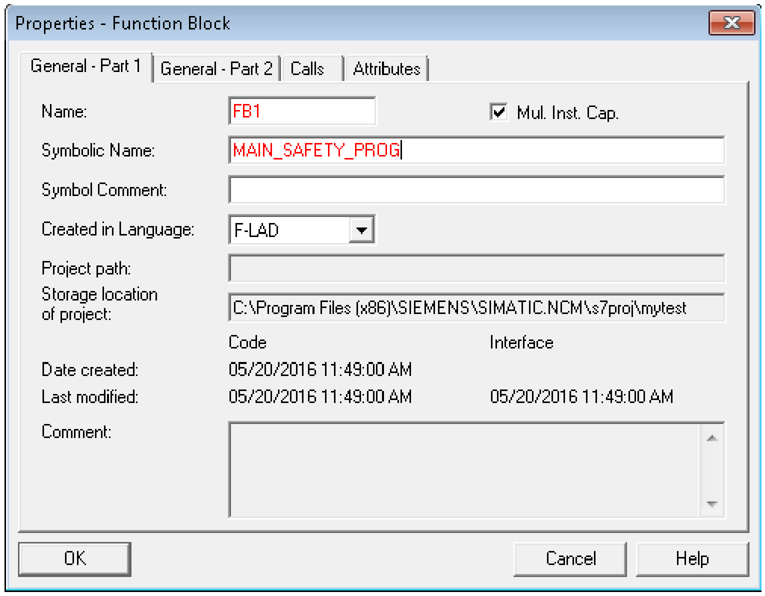
Now click **Insert** from menu and choose **S7 Block** and select appropriate block type to be inserted as shown in below figure.



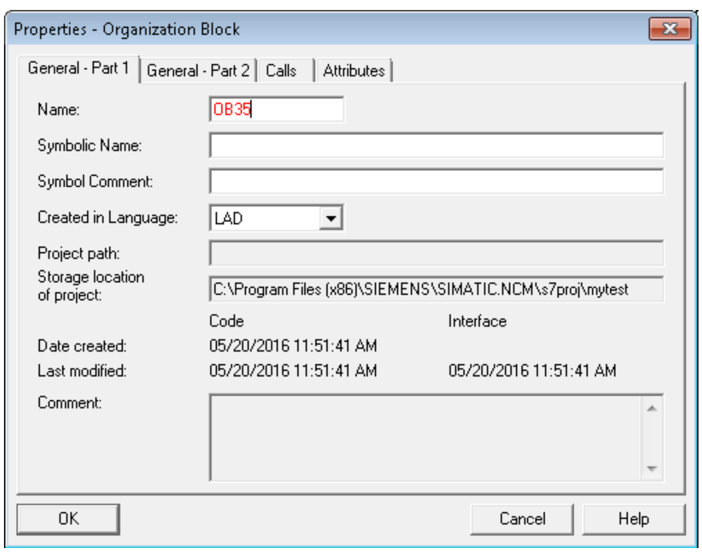
1. Insert 3. Function and select name and language as below.



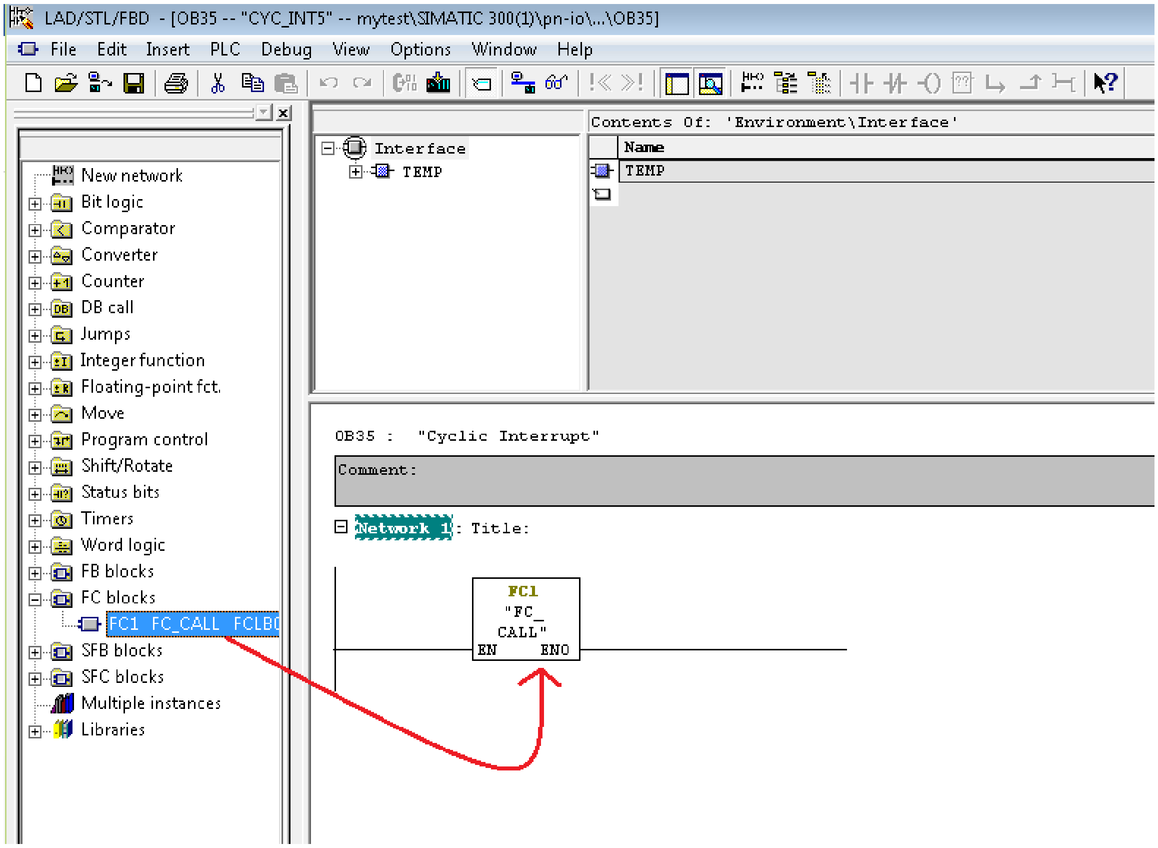
2. Insert 2. Function Block and select name and language as shown below.



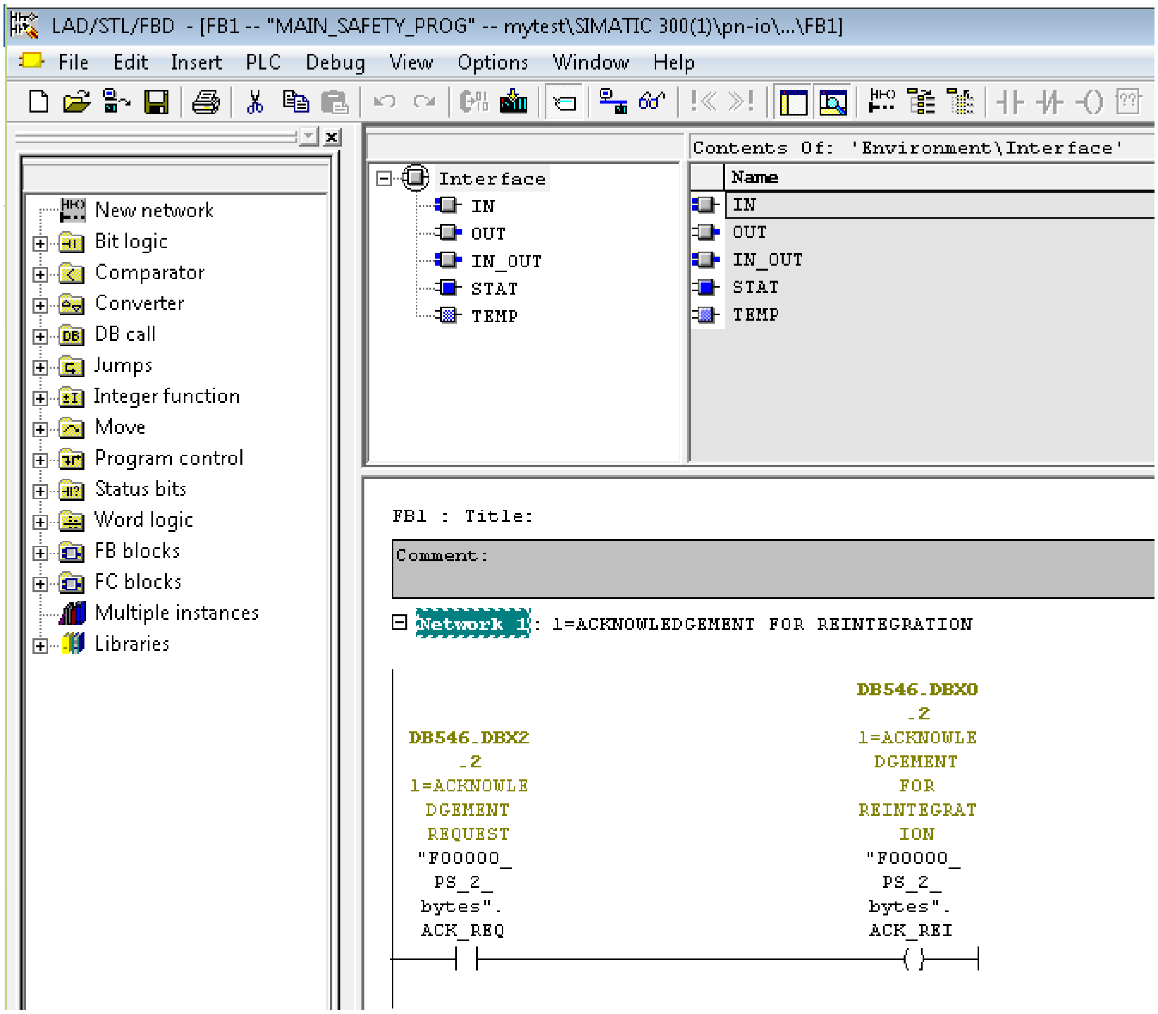
3. Insert 1. Organization Block and select name and language as shown below.



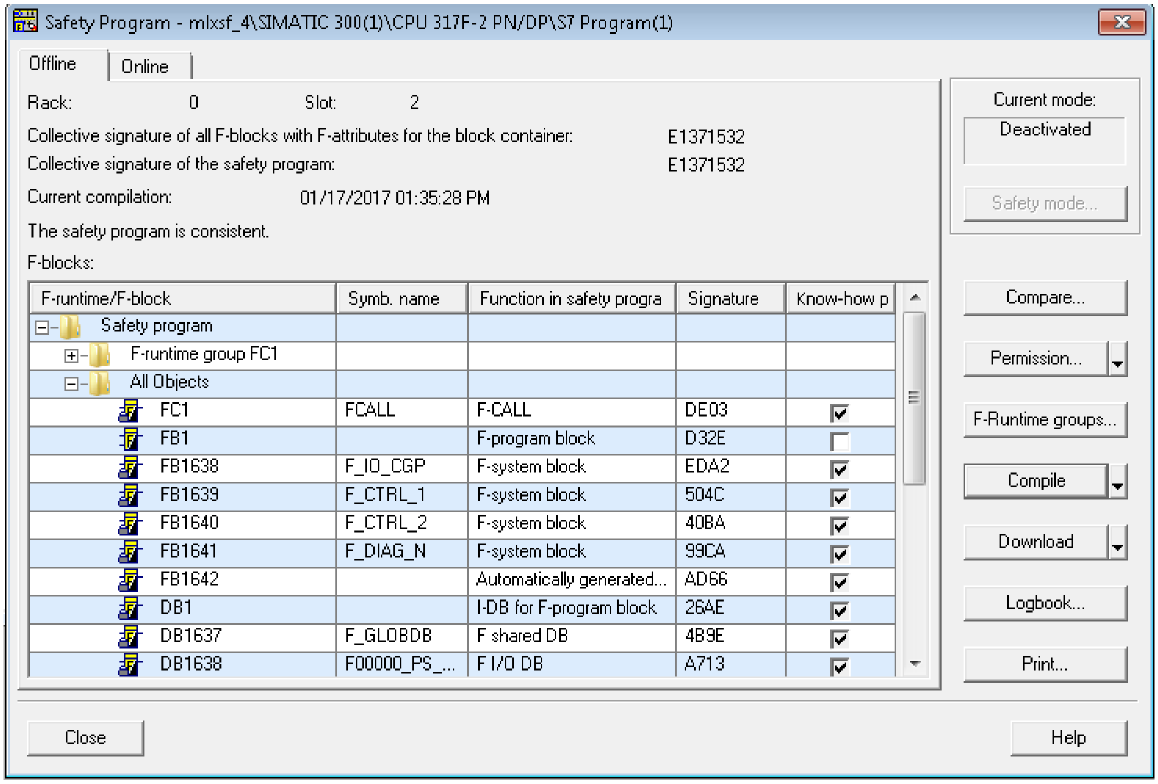
Now add FC_CALL to network on OB35. Open up OB35 by double clicking it. Then add F-CALL to Network by dragging from left pane to right window as shown below.



Now create reintegration logic. Click on FB1 (SAFETY BLOCK) to create this logic. In order to find ACK_REQ look for Data block that says PS X Bytes e.g. DB546 F00000_PS_2_Bytes. Add Examine ON to the Rung and set it to ACK_REQ i.e. DB546.DBX2.2. Add Output energizer to the rung and set it to ACK_REI i.e. DB456.DBX0.2.

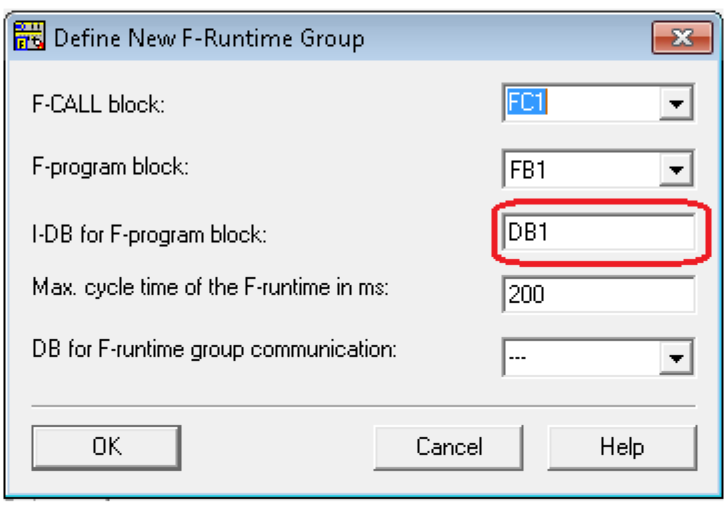


Go to SEMATIC Manager then Options ➔Edit Safety Program

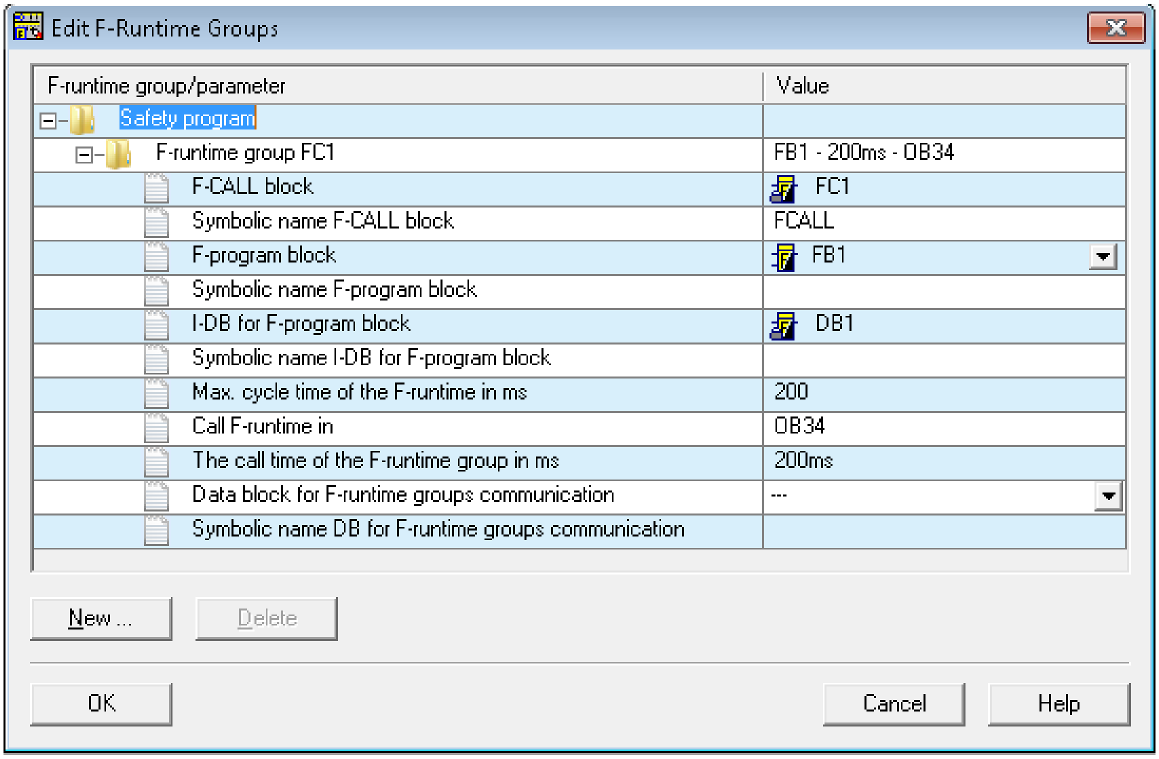


Now Click **F-Runtime group…** ➔New then one window pops up...

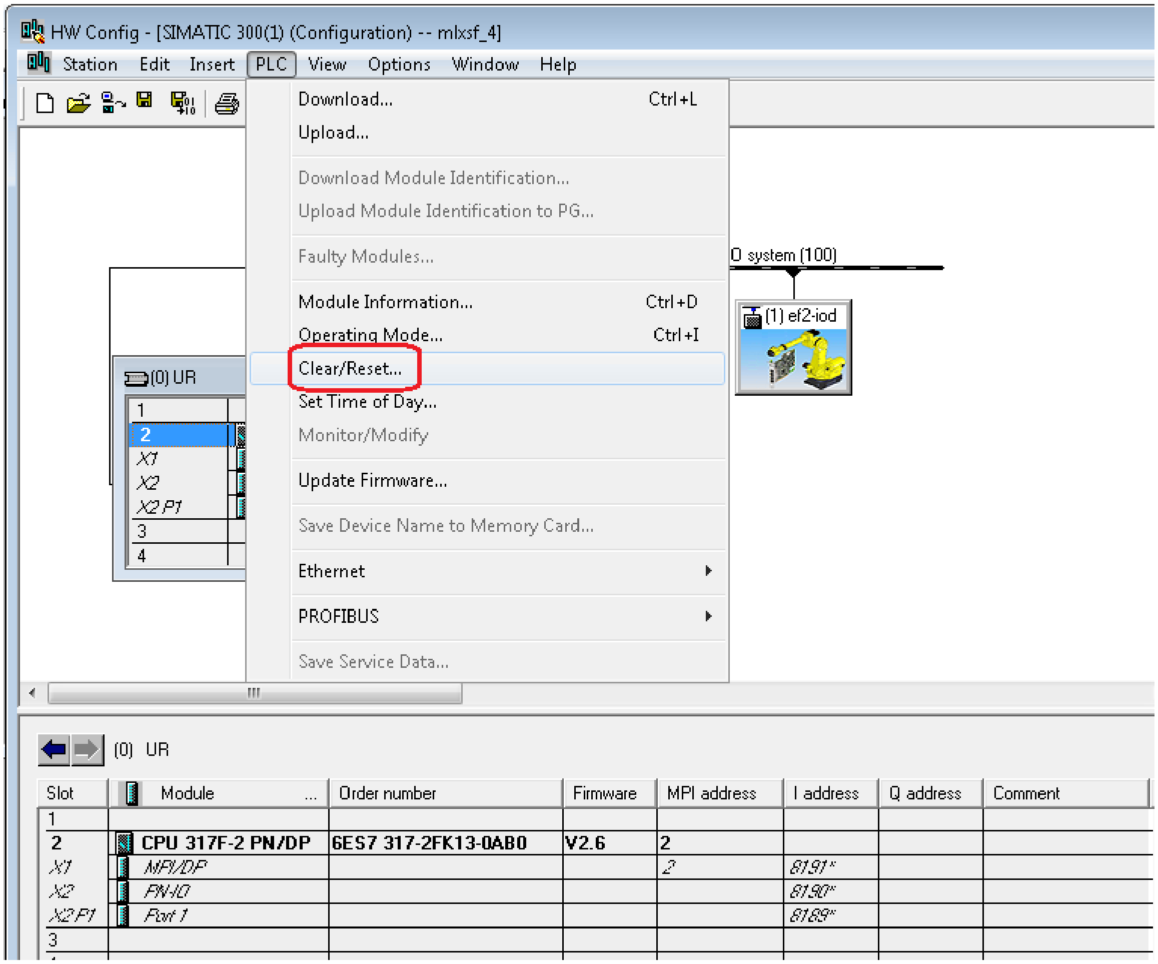
Type name in DB name, let say DB1 in this example.



Once you press OK, then it will create this instance block and link it to safety program. That should all blocks assigned properly as shown in figure. Now click OK. You will see window like below.



Click OK. Now you are back to Edit Safety Program window. Press compile, you may be asked for safety password which you configured previously in CPU properties/Protection tab. Hopefully everything is good at this point. Otherwise look for the errors and correct them. Once it compiles successfully, then download this to safety PLC and life should be good if everything went per plan. If safety did not incrementing frame counts in Robot, please try to RESET the memory and reload the program to PLC. Select the CPU in hardware config and from option drop down select CLEAR/RESET…



Robot Side Setup for IO-Device and Safety

Please note that Safety module MUST be first (Slot 2) after head module (Slot 1) as we did in STEP 7. Please note that in FW V2.6 (CP1604), slot # starts from 1 while FW V2.5 and below slot # starts from 0.

To setup with FW V2.6 :

Please use
"GSDML-V2.3-Fanuc-A05B2600J930V820D4-20131203.xml"
because I/O Device =Enabled, I/O Controller = Disabled.

Please renumber slot number from 1 because slot 0 can't be used with V2.6.

Please change the ModId from 0x18 to 0x1c (28 in decimal).

- Slot 1, subslot 1, ModID 0x1c
- Slot 2, subslot 1, ModID 0x27 (DO)

- Slot 3, subslot 1, ModID 0x28 (DI)

Just a note for V2.5 Device ID = 0x0002 and for V2.6 Device ID = 0x0008.

Please note that DI and DO order must be reversed as configured in PLC.

PROFINET V8.20 compatibility with current version

PROFINET(J930)/PROFINET Safety(J931) are under conformance test with V2.6 FW. and CP1604. The device ID must be changed and access points of V2.5 must be deleted from GSD file to pass the test. This means that V2.5 FW. can no longer be used when the new certified software is merged to V8.20.

This would bring a compatibility problem for the customer who is already using V2.5 FW. with V8.20. Updating software may need to upgrade the FW. to V2.6, and create STEP7 project again. Loading PNIO.SV from old version should be handled by new software.

As for PROFINET Safety(J931), using V2.6 FW. with CP1604 is necessary when the new certified software is merged to V8.20 because every safety function needs TUEV certification. This limitation can't be avoided.

On the other hand, FANUC consider making a compatibility flag (the bit0 of $PNIO_CFG.$CUSTOM) for the customer who insists using V2.5 firmware but doesn't care about conformance and doesn't use PROFIsafe. Using the flag is like using CP1616 where any conformance test isn't done.

Compatibility flag for V2.5 FW.

The customer sets the following system variables:

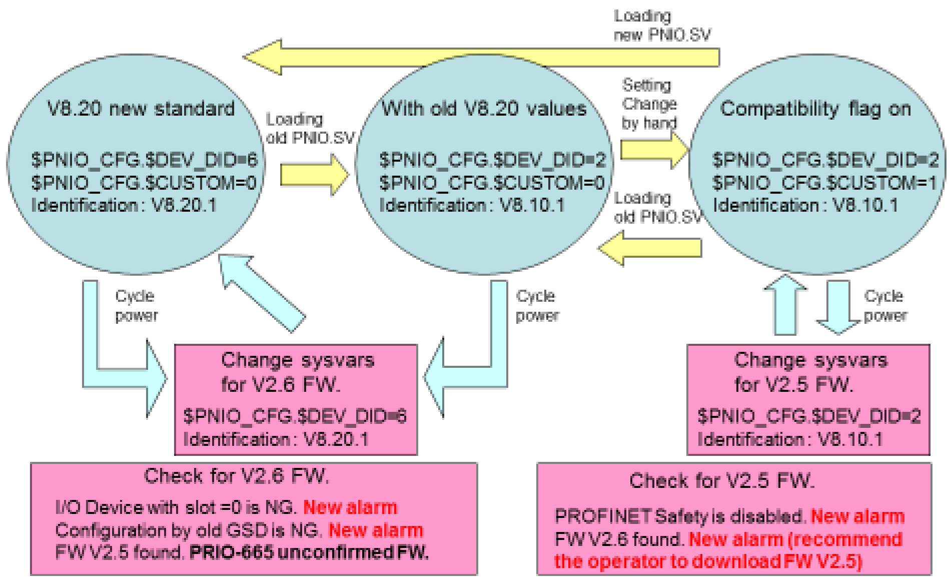
$PNIO_CFG.$CUSTOM = 1 (bit 0=1 is the compatibility flag)
$PNIO_CFG.$DEV_DID = 2 (current device ID for V8.20)
$PNIO_CFG2.$FW_VER_REQ[1] = 2
$PNIO_CFG2.$FW_VER_REQ[2] = 5
$PNIO_CFG2.$FW_VER_REQ[3] = 2
$PNIO_CFG2.$FW_VER_REQ[4] = 2
(V2.5.2.2 is the firmware version certified with V8.10)

PROFIsafe is disabled when the setting is used. New alarm will be added to notify this to operator.

New alarm with a message like "Downgrade FW. to V2.5" will be added, and it shows up when the setting is used but V2.6 firmware is found.

Note : The new device ID for V8.20 is 6. The device ID = 5 is reserved for V7.63.

Loading PNIO.SV and device ID



Note : The minimum slot number is changed from 0 to 1 in FW V2.6. Slot=0 can't be used for FW V2.6.

Procedure (STEP 7): Configure IO Modules

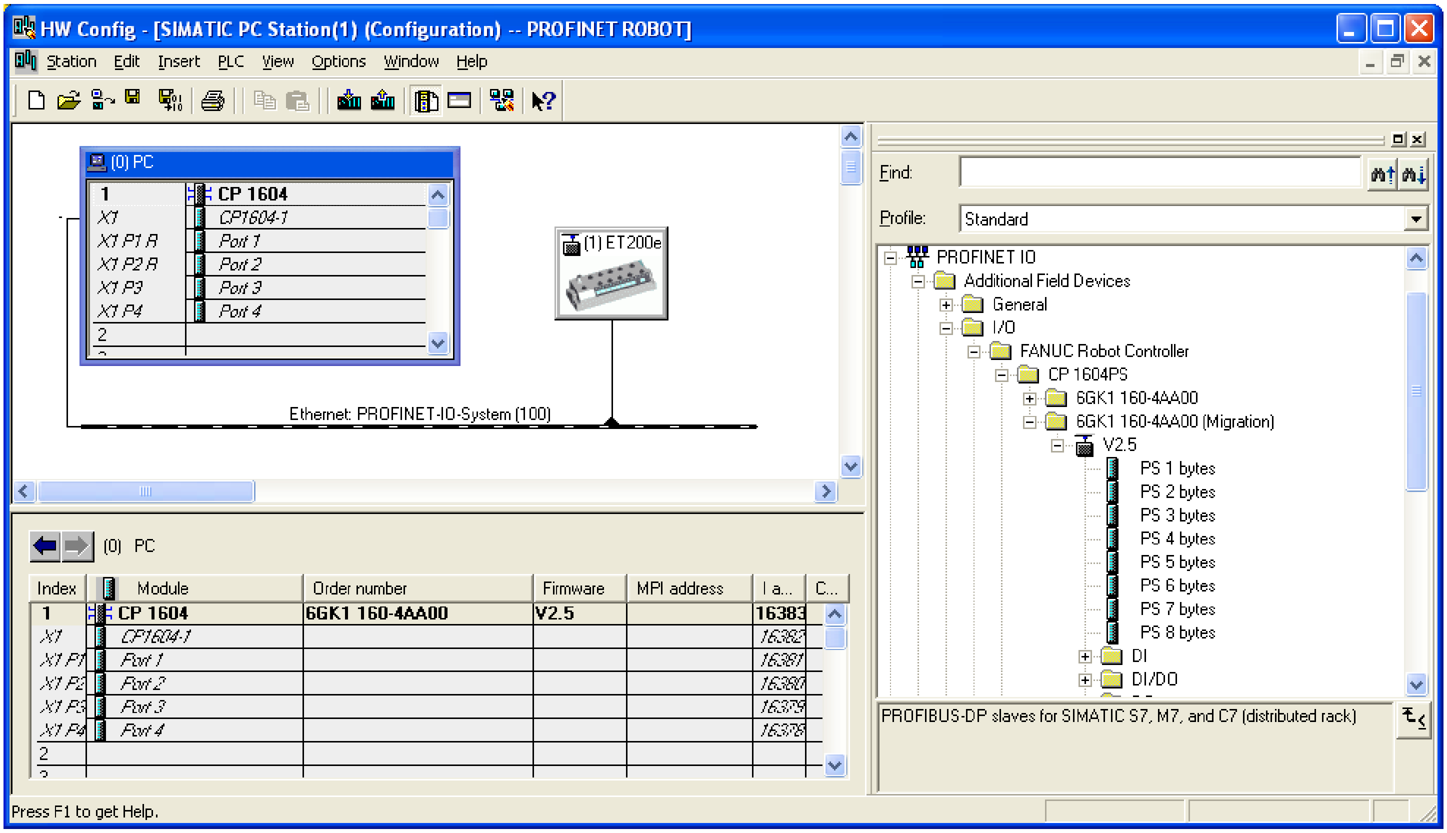
In this example, one 16 bytes DI module and one 16 bytes DO module are configured.

- ➢ Double click “CP1604” icon in NetPro window.
- ➢ Select slot 1 of CP1604 in the left-side down window.
- ➢ Select the following in the list in the right-side of HW Config window, and double click to insert in respective slot.

PROFINET I/O-➔I/O➔SIMATIC PC-CP➔CP1604➔V2.6➔DI/DO➔16 bytes

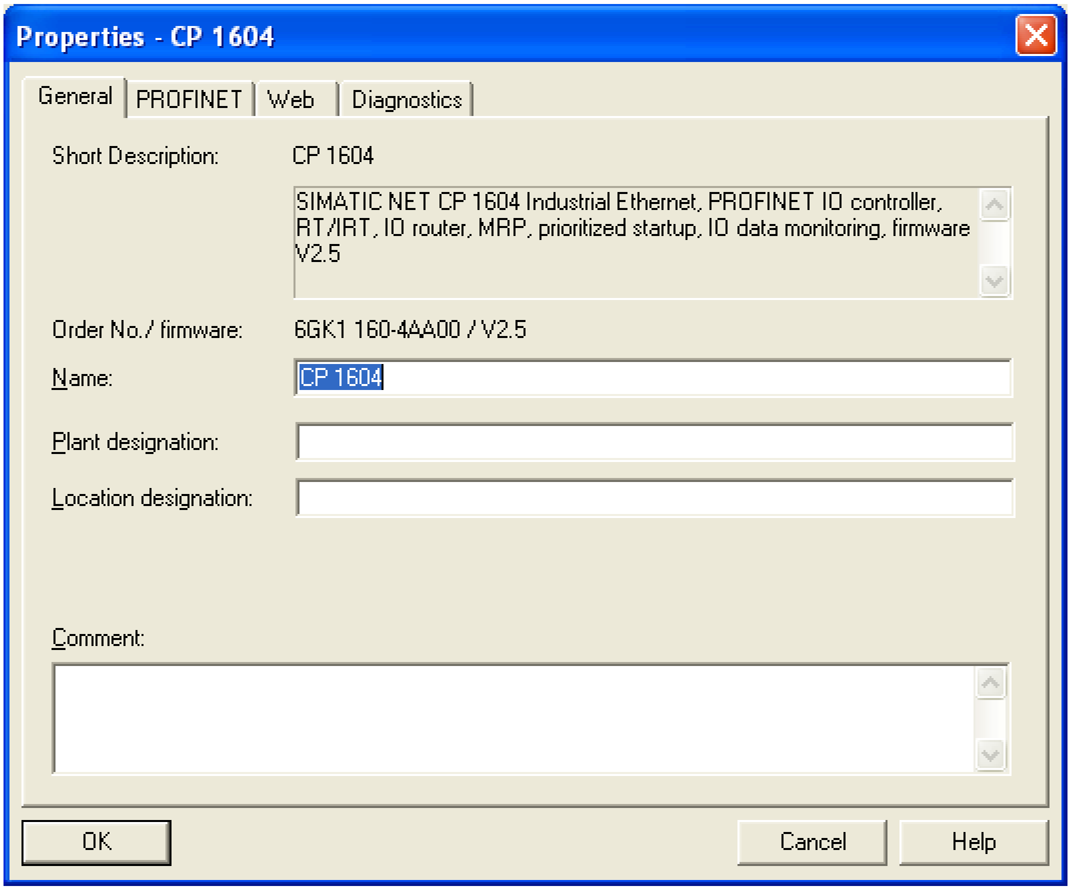
IO-Controller Configuration

Robot is IO-Controller and Molex block is IO-Device in below configuration screen. Add SIMATIC PC station to main project in order to do IO-Controller configuration. Now double click and add appropriate Molex (IO-Device) block as shown below.

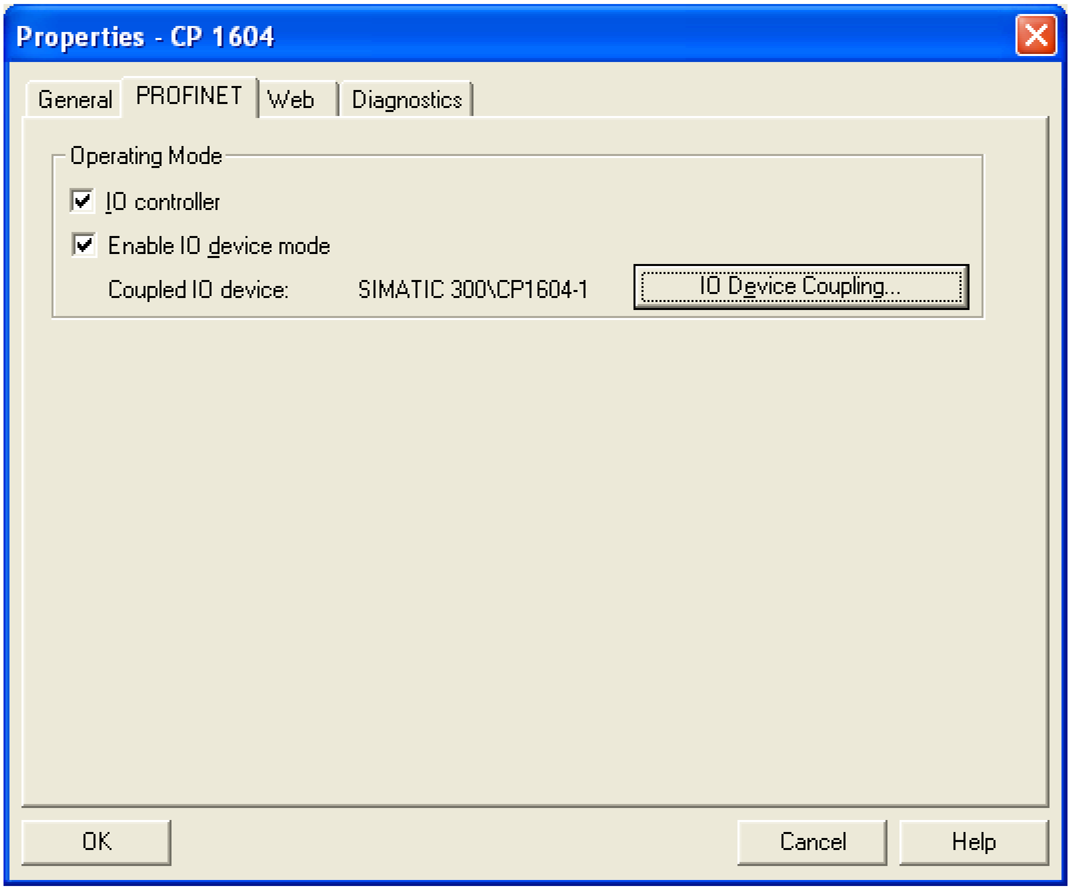


4. Properties of Modules

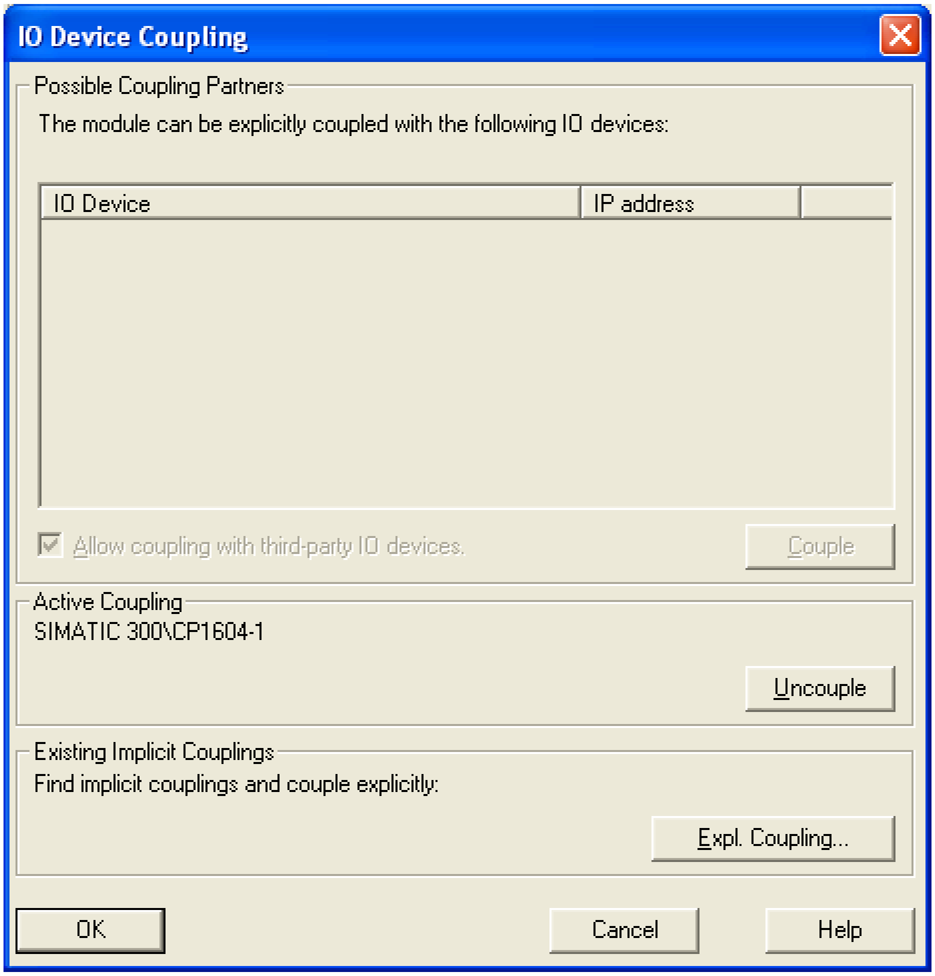
Double click on CP1604. General tab shows up device name and firmware information.



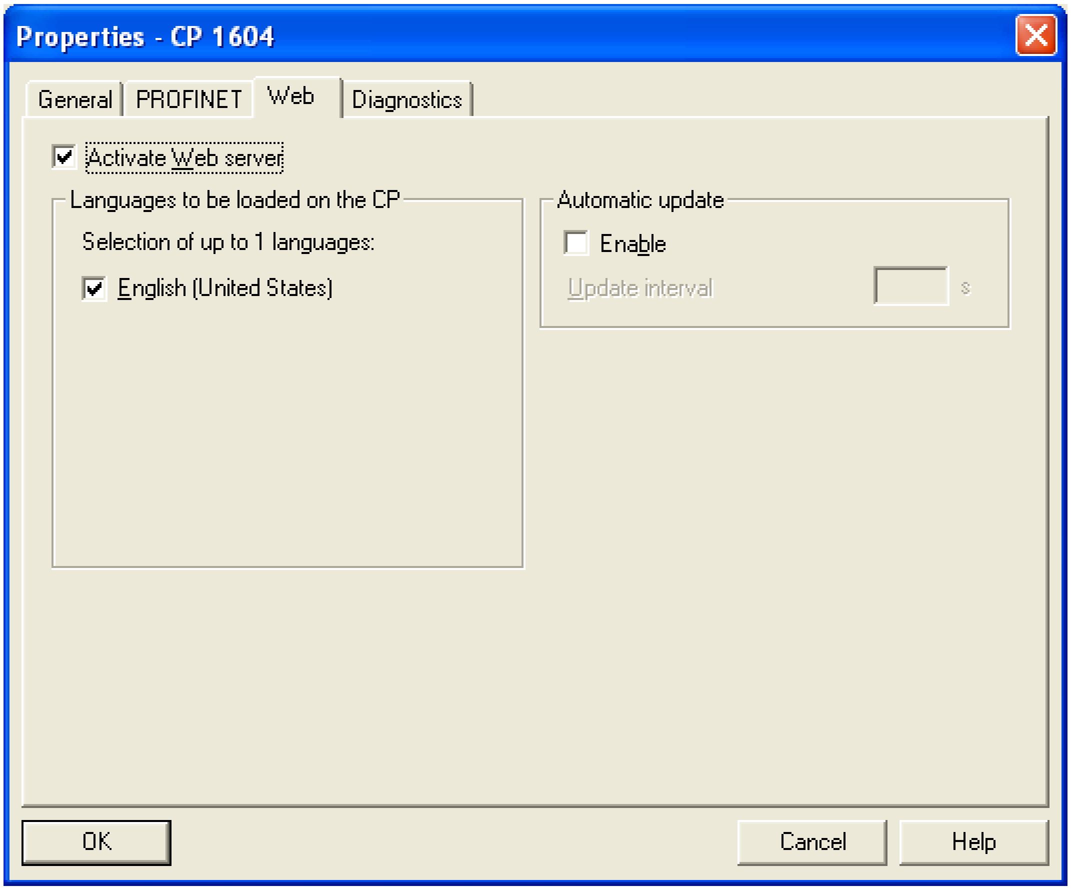
Click on PROFINET tab. Check both the check boxes: IO-Controller and Enable IO-Device mode.



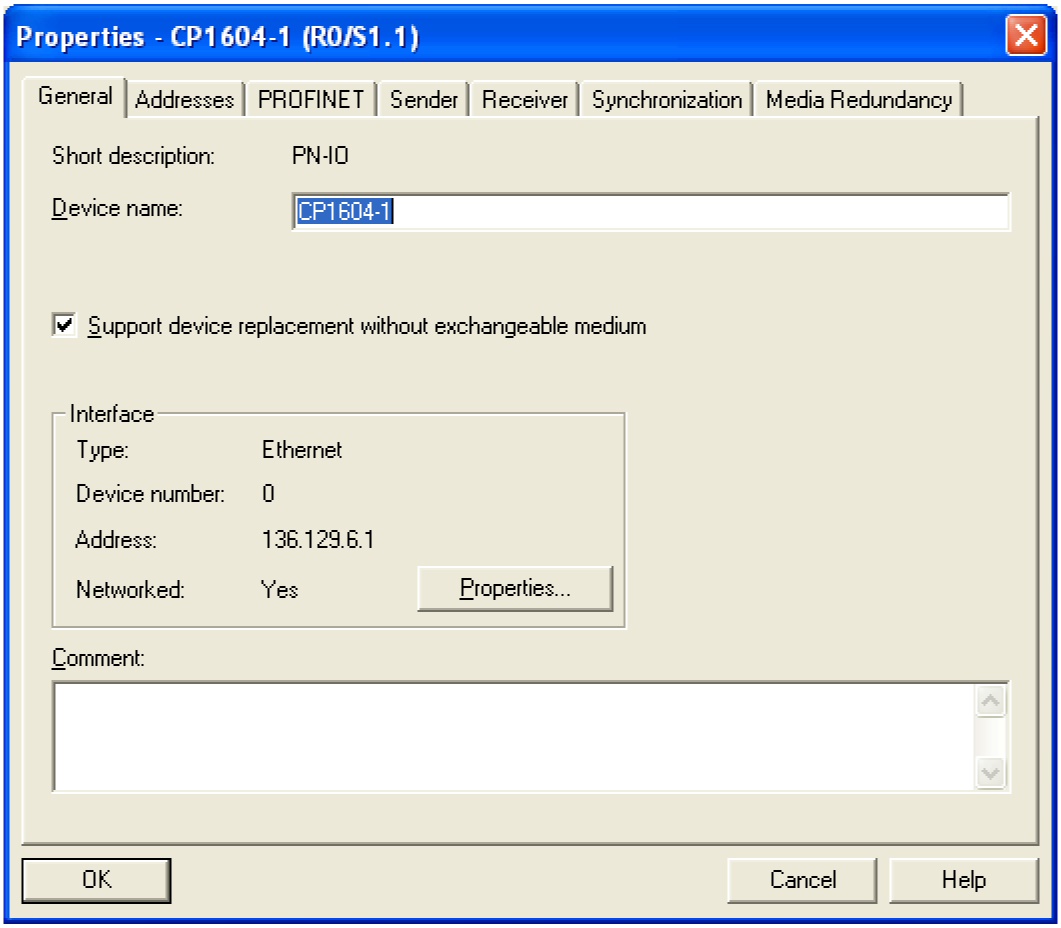
Click on IO Device coupling to enable simultaneous operation of IO-Device and IO-Controller on the same channel. Select the device and hit couple. If coupling is successful, you will see under active coupling.

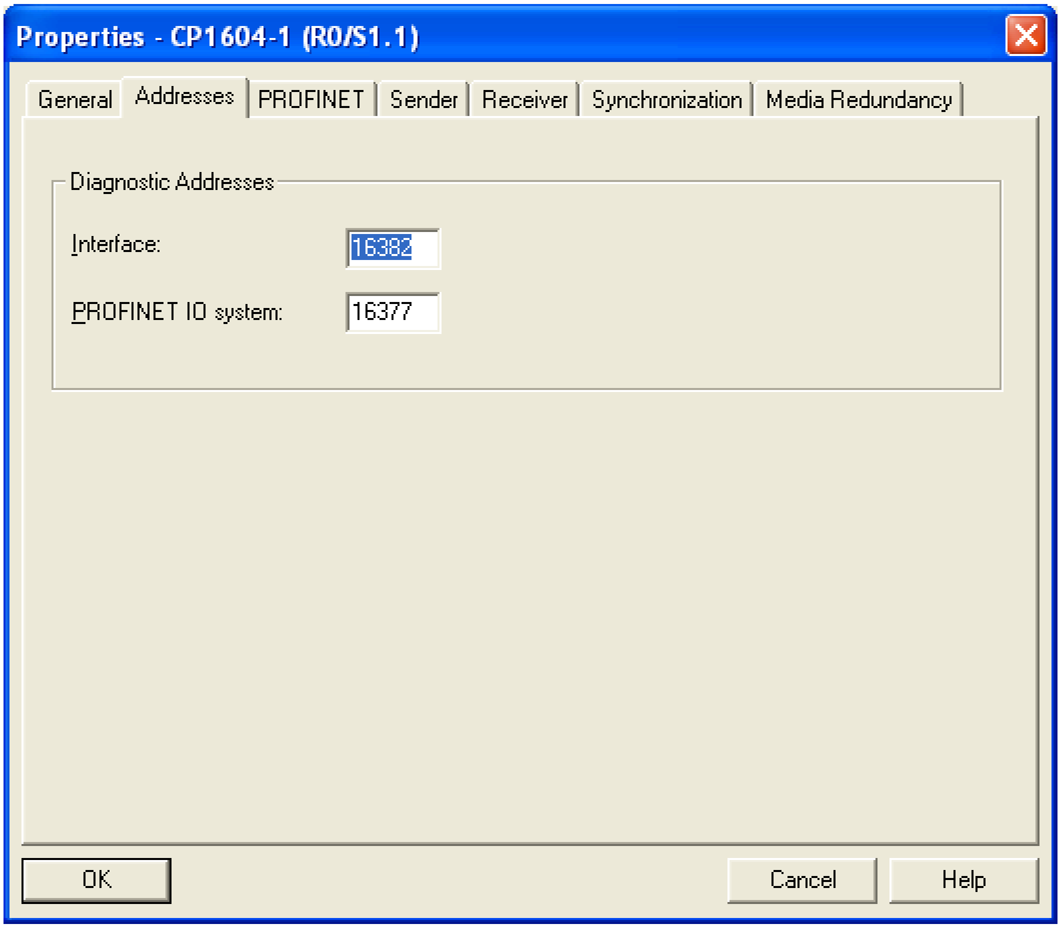


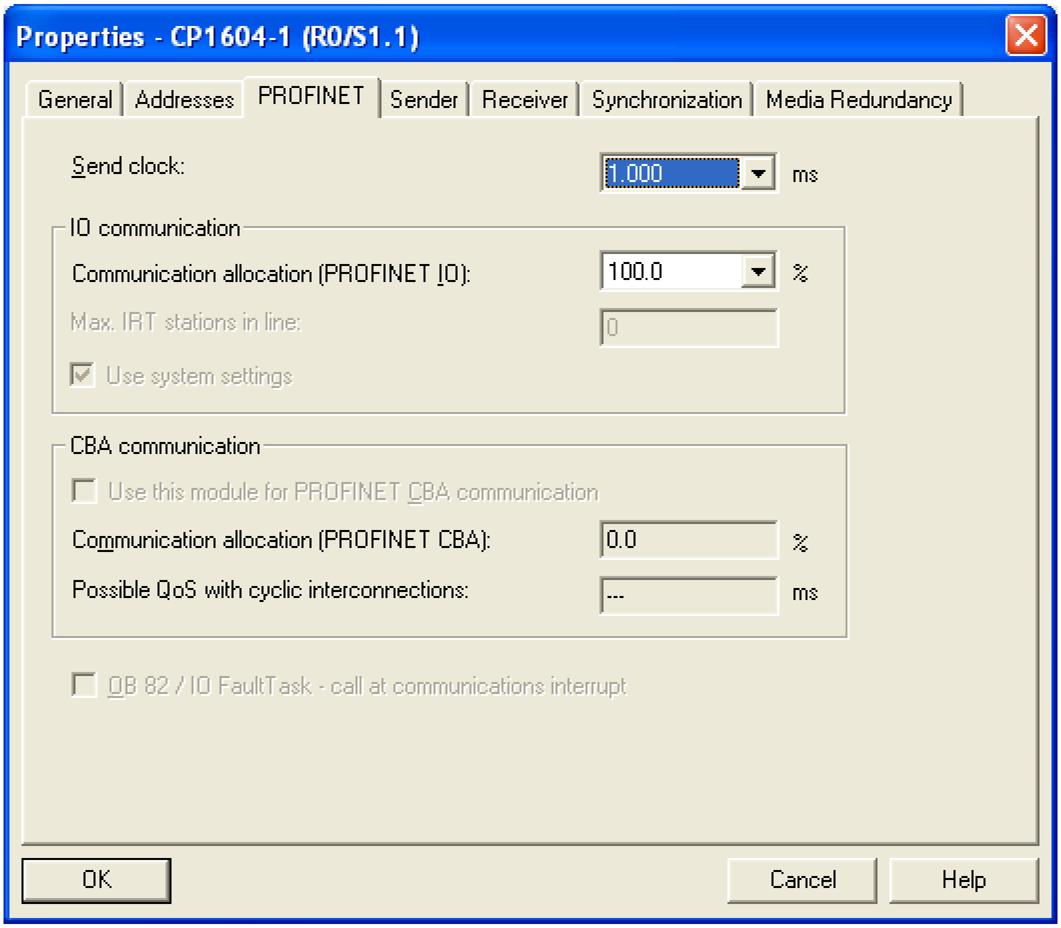
Go to web tab to enable web server and language support.

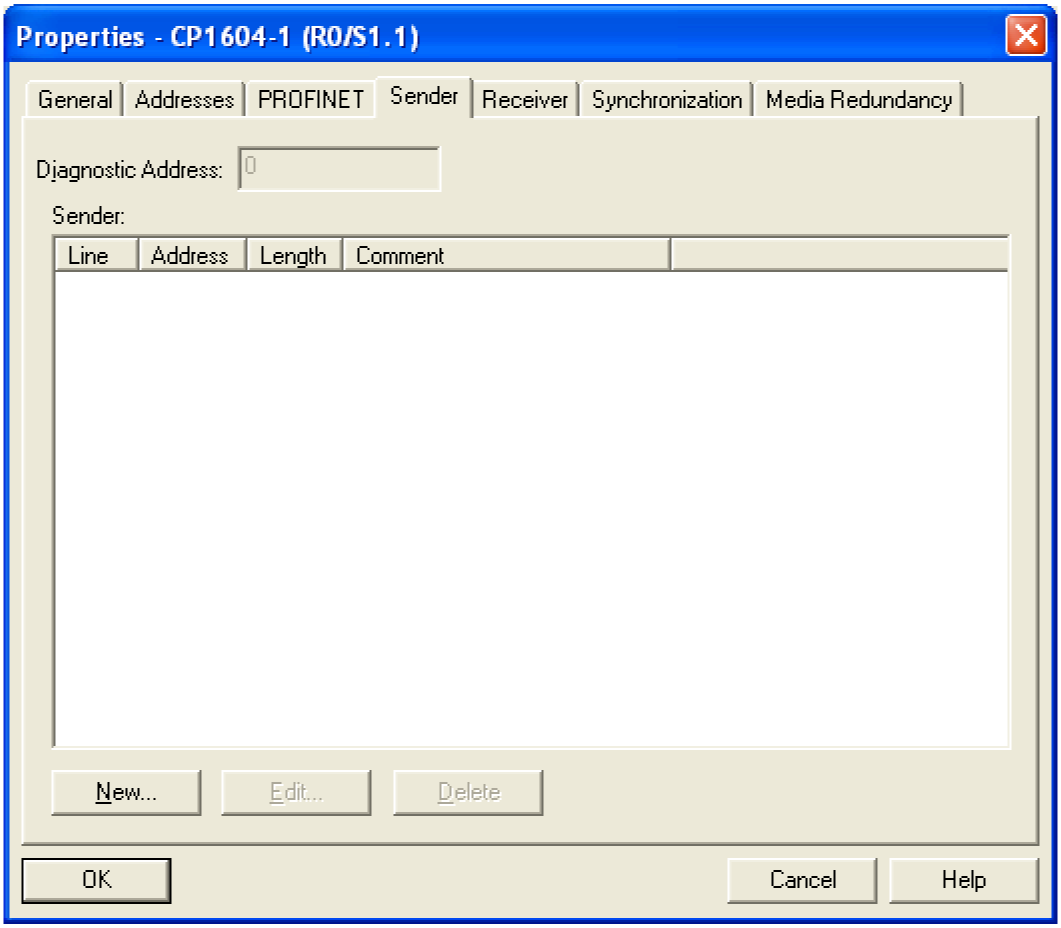


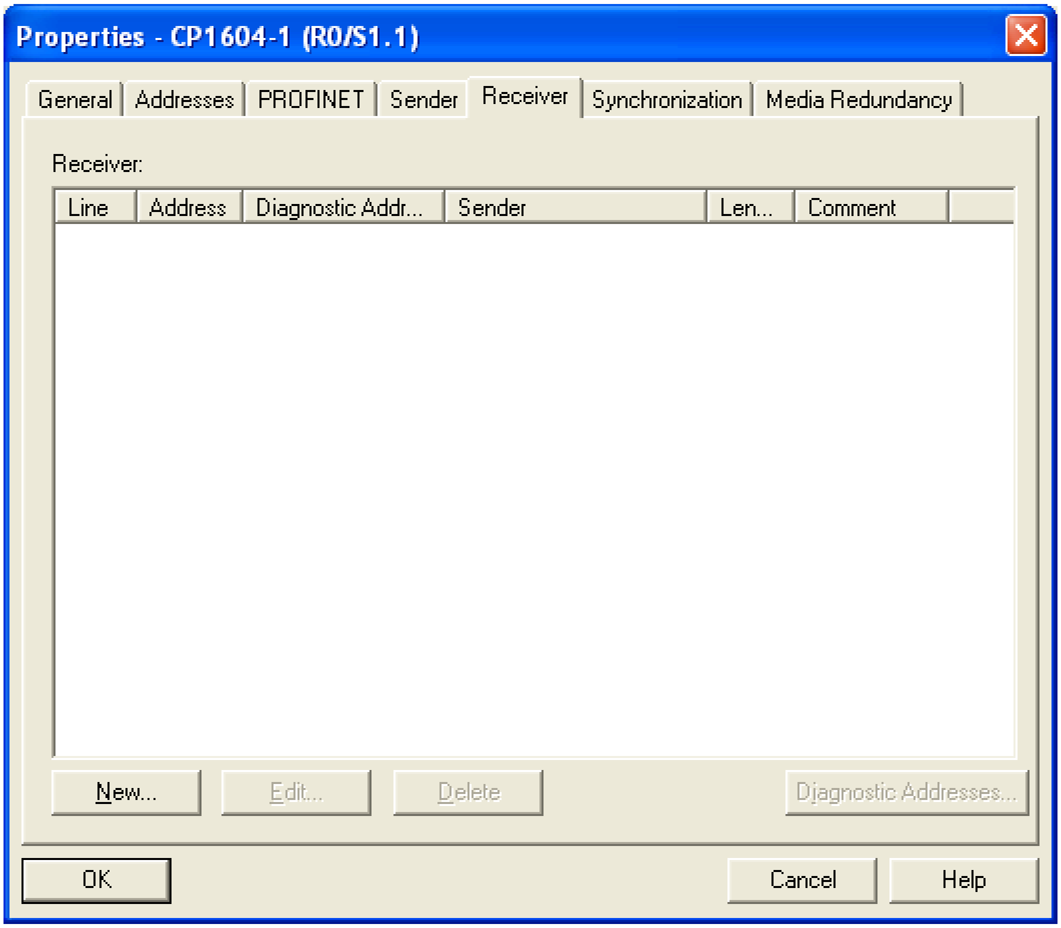
Double click on next module CP1604-01.

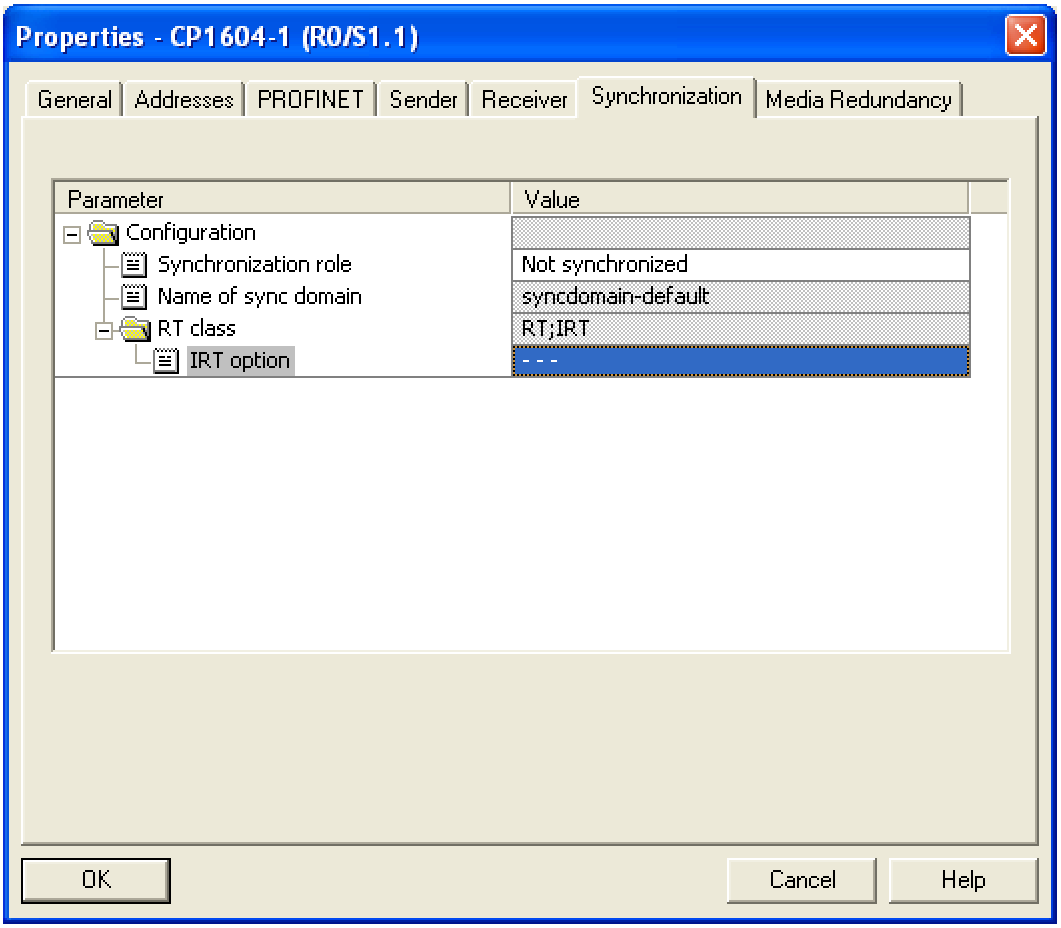

Properties - CP1604-1 (R0/S1.1)
General
Addresses
PROFINET
Sender
Receiver
Synchronization
Media Redundancy
Short description:
PN-IO
Device name:
CP1604-1
Support device replacement without exchangeable medium
Interface
Type:
Ethernet
Device number:
0
Address:
136.129.6.1
Networked:
Yes
Properties...
Comment:
OK
Cancel
Help

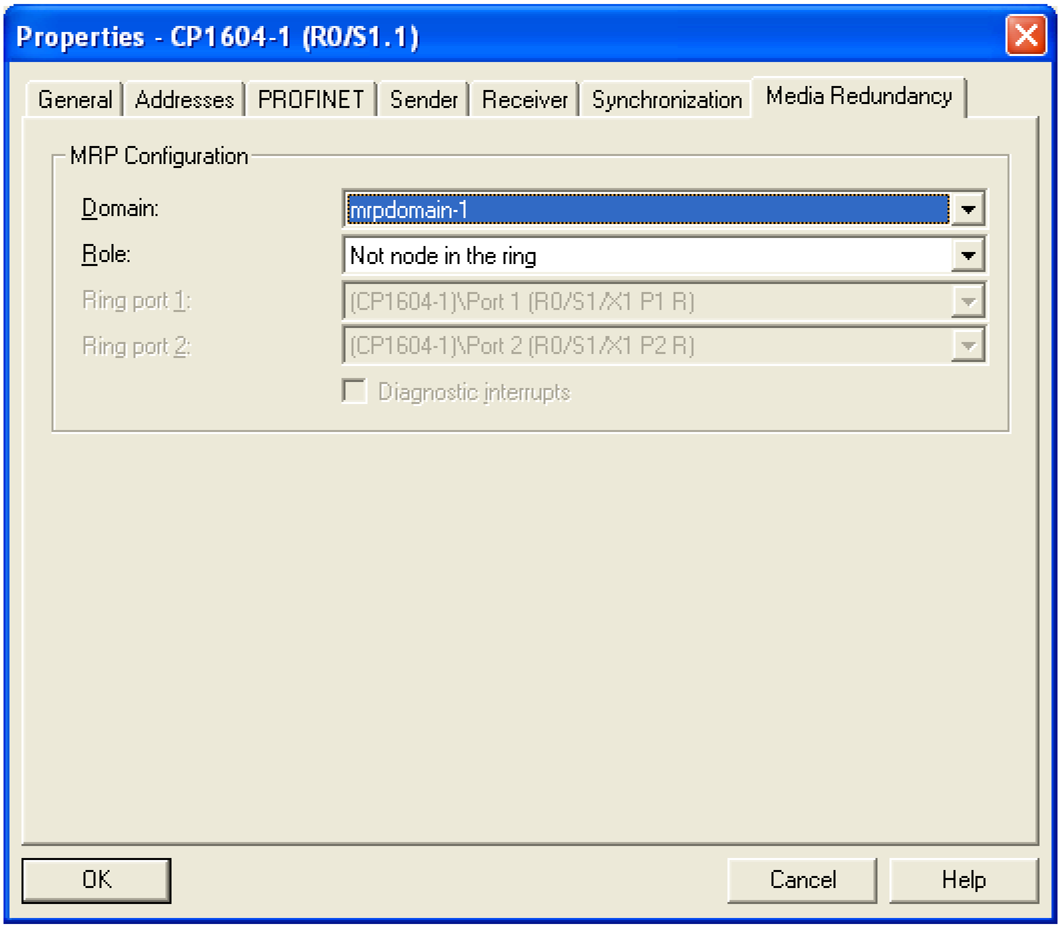

Properties - CP1604-1 (R0/S1.1)
General
Addresses
PROFINET
Sender
Receiver
Synchronization
Media Redundancy
Diagnostic Addresses
Interface:
16382
PROFINET IO system:
16377
OK
Cancel
Help


Properties - CP1604-1 (R0/S1.1)
General
Addresses
PROFINET
Sender
Receiver
Synchronization
Media Redundancy
Send clock:
1.000
ms
IO communication
Communication allocation (PROFINET IO):
100.0
%
Max. IRT stations in line:
0
Use system settings
CBA communication
Use this module for PROFINET CBA communication
Communication allocation (PROFINET CBA):
0.0
%
Possible QoS with cyclic interconnections:
---
ms
OB 82 / IO FaultTask - call at communications interrupt
OK
Cancel
Help


Properties - CP1604-1 (R0/S1.1)
General
Addresses
PROFINET
Sender
Receiver
Synchronization
Media Redundancy
Diagnostic Address:
0
Sender:
Line
Address
Length
Comment
New...
Edit...
Delete
OK
Cancel
Help


Properties - CP1604-1 (R0/S1.1)
General
Addresses
PROFINET
Sender
Receiver
Synchronization
Media Redundancy
Receiver:
Line
Address
Diagnostic Addr...
Sender
Len...
Comment
New...
Edit...
Delete
Diagnostic Addresses...
OK
Cancel
Help


Properties - CP1604-1 (R0/S1.1)
General
Addresses
PROFINET
Sender
Receiver
Synchronization
Media Redundancy
Parameter
Value
Configuration
Synchronization role
Not synchronized
Name of sync domain
syncdomain-default
RT class
RT;IRT
IRT option
- - -
OK
Cancel
Help



Compile and download to appropriate device.

Please note on IO-Controller on robot, you need to set correct Input and Output bits in IO-Controller screen (MENU➔SETUP➔PROFINET➔F4[CHOICE]➔IO-Controller).